

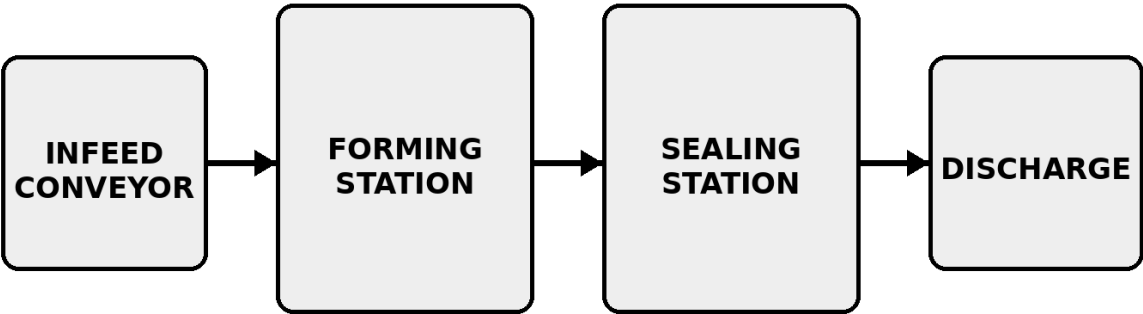
OPERATION AND MAINTENANCE MANUAL

AXP-420 Automatic Cartoning Line

For folded cartons, bottles, pouches, and preassembled product kits

Original instructions — English (United Kingdom)

AXP-420 AUTOMATIC CARTONING LINE — PROCESS FLOW



Sensors: B1 carton presence | B2 product confirmation | B3 discharge clear

Machine model	AXP-420	Serial number	AXP42-26-0178
Nominal output	Up to 120 cartons/min	Supply voltage	400 V AC, 3P+N+PE, 50 Hz
Pneumatic supply	6.0 bar, dry filtered air	Year of manufacture	2026
Manufacturer	Northstar Packaging Systems S.r.l.	Document revision	C — 30 July 2026

Document control

Revision	Date	Change description	Prepared by	Approved by
A	14 May 2026	Initial release for factory acceptance testing	M. Evans	R. Conti
B	22 June 2026	Added servo-homing and sanitation procedures	M. Evans	R. Conti
C	30 July 2026	Updated guarding interlock checks and alarm recovery	J. Patel	R. Conti

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- 3 Installation and commissioning

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1 Safety information

Read this manual before transporting, installing, operating, cleaning, adjusting, or servicing the machine. The AXP-420 contains moving chains, servo-driven mechanisms, pneumatic actuators, heated sealing components, and electrical circuits that can cause serious injury if energy sources are not isolated.

DANGER: Indicates an imminently hazardous situation that will result in death or serious injury if it is not avoided.

WARNING: Indicates a potentially hazardous situation that could result in death or serious injury.

CAUTION: Indicates a situation that could cause minor injury, product damage, or machine damage.

NOTICE: Identifies information required to prevent process interruption or loss of production data.

1.1 Intended use

The machine erects pre-glued folding cartons, inserts one verified product or product kit, closes the carton flaps, and transfers the finished carton to downstream equipment. Approved carton dimensions and product masses are listed in the format specification supplied with each change part set. Any use outside these limits requires written approval from Northstar Packaging Systems.

The following uses are prohibited:

- Processing explosive, unstable, radioactive, or biologically hazardous products.
- Operating with guards removed, interlocks defeated, or safety devices bypassed.
- Using compressed oxygen or combustible gas instead of industrial compressed air.
- Climbing on the conveyor, electrical cabinet, or machine frame.

1.2 Residual hazards

Hazard location	Hazard	Required control	Residual risk
Carton magazine	Crushing between follower and carton stack	Stop machine before replenishing below the minimum mark	Low
Main drive chain	Entanglement and drawing-in	Guard interlock; lockout before access	Low
Hot-melt option	Burns from nozzle and adhesive at 160–180 °C	Heat-resistant gloves; depressurise applicator	Medium
Pneumatic cylinders	Unexpected movement from stored pressure	Close and exhaust main air valve	Low
Electrical cabinet	Hazardous voltage up to 400 V AC	Qualified electrical personnel only	Low

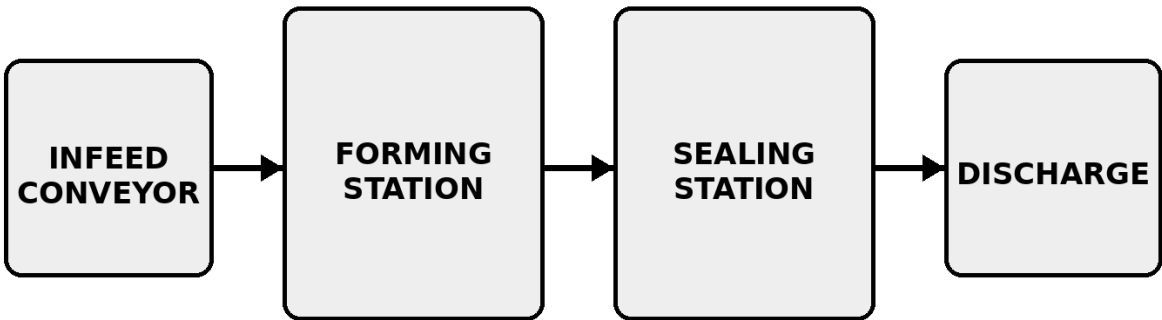
1.3 Lockout/tagout procedure

1. Stop production and remove all products and cartons from the machine.
2. Set the main isolator Q0 to OFF and secure it with a personal padlock.
3. Close pneumatic valve V0, lock it, and verify that pressure gauge PI0 reads 0 bar.
4. Wait at least five minutes for the servo drives and DC bus to discharge.
5. Try to start the machine from the HMI. No actuator shall move.
6. Perform the work, reinstall all guards, remove tools, and clear personnel before restoring energy.

2 Machine description

The AXP-420 is a modular horizontal cartoning machine. A servo-driven lug chain transports cartons through the loading and closing stations. Product presence is confirmed before closure; rejected cartons are diverted automatically without stopping the line.

AXP-420 AUTOMATIC CARTONING LINE — PROCESS FLOW



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Figure 1 — Main process stations and product flow

2.1 Main assemblies

Item	Assembly	Function
1	Carton magazine	Stores flat carton blanks and maintains controlled pressure against the pick-up position.
2	Vacuum pick-and-erect unit	Removes one carton blank, opens it, and places it between transport lugs.
3	Product loading station	Receives product from upstream equipment and confirms correct insertion.
4	Flap closing station	Folds dust flaps and closes the major flaps by tuck closure or hot-melt adhesive.
5	Reject station	Diverts incomplete or unverified cartons into a lockable reject bin.
6	HMI and control cabinet	Provides recipe control, diagnostics, safety monitoring, and production counters.

2.2 Operating modes

Automatic: Continuous production using the selected recipe. The safety circuit and all permissive conditions must be satisfied.

Manual/Jog: Individual mechanisms can be moved at reduced speed for setup and maintenance. Hold-to-run commands are required.

Setup: Allows format parameters to be adjusted and saved by authorised users.

Cleaning: Disables automatic product feeding and provides controlled conveyor movement for cleaning access.

2.3 Identification and terminology *Throughout this manual, operator side means the side containing the HMI. Drive side means the opposite side. Directions such as upstream and downstream refer to normal product flow.*

Support portal: service.northstar-packaging.example — use the machine serial number when requesting assistance.

3 Installation and commissioning

Installation shall be performed by qualified personnel using the approved layout drawing, foundation plan, and electrical schematic. Store the machine indoors between -10°C and $+45^{\circ}\text{C}$. Relative humidity shall remain below 85%, non-condensing.

3.1 Site requirements

Requirement	Minimum	Recommended	Notes
Clear floor area	6.8 m × 2.4 m	7.5 m × 3.0 m	Includes service access, not upstream/downstream equipment.
Floor load capacity	8 kN/m ²	10 kN/m ²	Machine mass is approximately 2 450 kg.
Electrical supply	400 V $\pm 10\%$, 50 Hz	Dedicated 32 A feeder	TN-S system; prospective short-circuit current ≤ 10 kA.
Compressed air	5.5 bar at 800 NI/min	6.0 bar at 1 000 NI/min	ISO 8573-1 class 4:4:3 or better.
Ambient temperature	+5 to $+40^{\circ}\text{C}$	$+18$ to $+25^{\circ}\text{C}$	Avoid direct sunlight on optical sensors.

3.2 Mechanical installation

- Position the machine according to drawing GA-AXP420-001. Maintain at least 800 mm clearance at removable guards and 1 000 mm in front of the electrical cabinet.
- Level the main frame using the six adjustable feet. Maximum permitted deviation is 0.5 mm per metre in both axes.
- Install the supplied floor anchors only after alignment with upstream and downstream conveyors is complete.
- Connect product-transfer bridges without creating a trapping point or a vertical step greater than 1.5 mm.

3.3 Electrical and pneumatic connection

Route the incoming power cable through gland X0 and terminate conductors at Q0 according to the schematic. Protective earth must be connected before any live conductor. Verify clockwise phase sequence at the main fan and vacuum pump. Connect dry compressed air to the G 1/2 inlet on service unit FRL0; do not apply thread-sealing tape downstream of the filter.

3.4 Commissioning checklist

Check	Acceptance criterion	Recorded result
Guard interlocks	Opening each interlocked guard removes torque and displays the correct guard number.	Pass / Fail
Emergency stops	Every emergency-stop device stops hazardous motion and requires manual reset.	Pass / Fail
Servo homing	All axes reach home without mechanical contact or following error.	Pass / Fail
Vacuum system	Vacuum level is at least -0.65 bar during carton pick-up.	_____ bar
Reject verification	Test carton is rejected and reject counter increments by one.	Pass / Fail

Service note —After transport or relocation, inspect all plug connectors, pneumatic tubes, guards, and mechanical fasteners before energising the machine.

4 Operating instructions

4.1 Pre-start inspection

- All guards, covers, and access doors are installed and closed.
- The machine and surrounding floor are free of tools, loose fasteners, packaging debris, and cleaning materials.
- The carton magazine, product infeed, and reject bin contain the correct materials for the selected recipe.
- Air pressure is between 5.5 and 6.5 bar and no significant leakage is audible.
- No person is inside the guarded area or working on connected equipment.

4.2 Starting automatic production

11. Turn main isolator Q0 to ON and wait for the HMI login screen.
12. Log in with an authorised operator account.
13. Select the required recipe and compare the displayed carton code, product code, and change-part number with the production order.
14. Press RESET. The safety status banner must change from red to grey.
15. Press HOME and wait until the message “Machine homed — ready” appears.
16. Start upstream product supply, then press START. Increase speed gradually while observing carton erection and product insertion.

4.3 Recipe parameters

Parameter	Typical value	Allowed range	Unit	Effect
Machine speed	90	20–120	cartons/min	Sets the master production rate.
Carton pick vacuum delay	45	20–120	ms	Delays vacuum confirmation after cup contact.
Product insertion position	286.5	250.0–320.0	mm	Defines servo insertion endpoint.
Glue temperature	170	150–185	°C	Hot-melt option only.
Reject pulse duration	220	100–500	ms	Controls reject-cylinder extension time.

Do not compensate for worn tooling by entering values outside the validated recipe range. Excessive position offsets can cause collisions between the insertion pusher, carton lugs, and closing guides.

4.4 Format changeover

A complete changeover requires the matching colour-coded change-part kit. Parts marked blue are installed on the operator side; parts marked black are installed on the drive side. The HMI wizard shows each required part and its installation position.

17. Stop the machine at an empty-cycle position and apply lockout/tagout.
18. Remove the carton magazine guides, transport lugs, product pusher head, and closing rails specified by the current recipe.
19. Clean all mounting faces. Never install a change part over carton dust, adhesive residue, or a raised locating pin.
20. Install the new parts and tighten hand knobs firmly. Use a torque wrench only where a torque value is engraved.
21. Restore energy, select the new recipe, home all axes, and run five empty cartons at 20 cartons/min.
22. Introduce product and inspect ten consecutive finished cartons before releasing the line for production.

4.5 Product quality checks

Inspection point	Method	Acceptance
Carton closure	Visual inspection of top and bottom flaps	Flaps fully engaged; no tearing or adhesive squeeze-out
Coding and traceability	Verify printed lot code and camera result	Acceptance by market:

		EU	Lot + expiry date
		US	Lot + use-by date
		Export	Customer-specific code

5 Preventive maintenance

Maintenance intervals are based on two-shift operation in a clean packaging environment. Shorten the intervals where carton dust, adhesive vapour, wash-down chemicals, or continuous three-shift production are present.

Interval	Component	Task	Material/tool	Responsible person
Every shift	Machine interior	Remove carton dust with an industrial vacuum. Do not use compressed air near sensors or bearings.	Vacuum; lint-free cloth	Operator
Weekly	Vacuum cups and hoses	Inspect for cracks, flattening, blockage, and loose fittings.	Mild detergent solution	Technician
Weekly	Transport chain	Check tension and wipe guide surfaces.	Feeler gauge	Technician
Monthly	Safety devices	Function-test emergency stops, interlocks, and reset logic.	Test record FRM-SAF-02	Qualified technician
Every 1 000 h	Gearboxes	Inspect for leakage and abnormal noise; verify mounting bolts.	Torque wrench	Maintenance
Every 2 000 h	Servo coupling	Inspect elastomer insert and shaft alignment.	Dial indicator	Maintenance
Annually	Electrical cabinet	Inspect terminals, fans, filters, and protective-earth continuity.	Calibrated test instruments	Qualified electrician

5.1 Lubrication

Use only NSF H1 food-grade lubricants where specified. Mixing incompatible greases can cause bearing failure. Wipe each grease nipple before connecting the grease gun and remove excess lubricant after application.

Lubrication point	Lubricant	Quantity	Interval	Access condition
Carton transport chain	ISO VG 68 H1 chain oil	Thin continuous film	250 h	Machine stopped
Main cam bearings	NLGI 2 H1 grease	2 strokes per nipple	1 000 h	Locked out
Magazine linear guides	PTFE dry-film lubricant	Light spray, wipe dry	500 h	Locked out
Pneumatic service unit	None	Do not add oil	—	Factory dry-air design

5.2 Cleaning procedure

The standard AXP-420 is not designed for hose-down cleaning. Protect the HMI, electrical cabinet, motors, sensors, and pneumatic components from direct liquid spray. Use a damp cloth with neutral detergent on painted and stainless-steel surfaces. Do not use chlorine-based cleaners, abrasive pads, ketones, or pressurised steam.

6 Troubleshooting and alarm recovery

When an alarm occurs, note the alarm number and first-out event before resetting. Repeated resetting without correcting the cause can damage cartons, tooling, or servo mechanisms.

Alarm	HMI message	Probable cause	Diagnostic check	Corrective action	Restart condition
A101	Carton not picked	Magazine empty; vacuum leak; double blank	Check carton level, vacuum switch PS1, cup condition, and blank separation.	Refill magazine; clean or replace cups; adjust separator fingers.	One carton is erected correctly in manual mode.
A118	Product missing at loading station	Upstream gap; sensor B2 dirty; product stopped in transfer	Observe B2 status and inspect transfer guides.	Clean B2; remove obstruction; synchronise upstream equipment.	Product presence changes reliably on HMI.
A205	Insertion axis following error	Mechanical obstruction; excessive acceleration; loose coupling	Lock out and move mechanism by hand; inspect coupling and guides.	Remove obstruction; tighten coupling; restore validated motion profile.	Axis homes without error and completes five dry cycles.
A233	Guard 3 open	Guard not closed; actuator misaligned; cable fault	Inspect guard tongue and safety-switch LEDs.	Close guard; realign actuator; repair cable using approved parts.	Safety controller shows both channels healthy.
A310	Low air pressure	Plant pressure low; V0 closed; filter blocked; major leak	Read PI0 and listen for leakage.	Open V0; replace filter element; repair leaking tube or valve.	Pressure remains above 5.5 bar during cycling.
A402	Reject bin full or absent	Bin full; bin not seated; sensor B9 misaligned	Inspect bin and B9 indication.	Empty and reinstall bin; align sensor.	B9 confirms bin present and not full.
A510	Glue temperature not ready	Heater warming; sensor fault; overtemperature trip	Check actual temperature, setpoint, and applicator controller status.	Wait for ready; inspect sensor; reset controller after cooling.	Actual temperature is within ± 5 °C of setpoint.
A612	Downstream blocked	Discharge photoeye B3 covered; downstream conveyor stopped	Inspect discharge and downstream permissive input.	Remove cartons; restart downstream conveyor; clean B3.	Discharge path remains clear for three seconds.

7 Technical data

Characteristic	Standard machine	Optional configuration	Tolerance/remark
Maximum mechanical speed	120 cartons/min	150 cartons/min high-speed kit	Product and carton dependent
Carton length	80–240 mm	60–300 mm extended range	Inside dimension
Carton width	35–110 mm	25–140 mm	Inside dimension
Carton height	18–85 mm	15–110 mm	Inside dimension
Installed electrical power	12.5 kW	18.0 kW with hot melt	Actual demand typically 4–8 kW
Sound pressure level	76 dB(A)	79 dB(A) with vacuum pump enclosure open	Measured at operator position
Machine mass	Approximately 2 450 kg	Up to 2 850 kg	Without separate infeed modules
Control voltage	24 V DC	—	PELV power supply

7.1 Pneumatic schematic notes

The main air preparation unit consists of manual shut-off valve V0, lockable exhaust valve, filter F0, regulator PR0, and pressure switch PS0. The vacuum generator uses a separate branch with silencer SL1. Tubes are identified by function and port number, for example “YV12-2 → CY12-A”. Do not swap cylinder ports to reverse motion; change direction only through the approved control configuration.

7.2 Electrical reference designations

Prefix	Meaning	Example
B	Sensor	B2 product-present photoelectric sensor
K	Relay or contactor	K1 main control contactor
M	Motor	M4 vacuum pump motor
Q	Switch or circuit breaker	Q0 main isolator
S	Operator control or safety device	S0 emergency-stop pushbutton
X	Terminal or connector	X14 downstream interface connector
YV	Solenoid valve	YV12 reject-cylinder valve

8 Spare parts and service information

Use original spare parts or parts approved in writing by Northstar Packaging Systems. When ordering, provide the machine model, serial number, assembly drawing number, item number, description, and required quantity. Safety-related components must have the same manufacturer, type, contact arrangement, and performance level as the original component unless an approved engineering change is supplied.

Part number	Description	Recommended stock	Criticality	Storage note
NS-VC-32-SI	Silicone vacuum cup, Ø32 mm	12	Production	Store flat, away from UV light.
NS-BELT-AT5-825	Timing belt, AT5, 825 mm	2	Production	Do not bend below 80 mm radius.
NS-SEN-PH-M18	Photoelectric sensor, M18, PNP	2	Production	Keep connector cap installed.
NS-SAFE-RFID-01	Coded RFID guard switch	1	Safety	Replacement requires validation test.
NS-FILT-40UM	Pneumatic filter element, 40 µm	4	Maintenance	Keep sealed and dry.

9 Translation and terminology notes

The following terms are intentionally used consistently throughout the manual: “guard” means a physical protective barrier; “interlock” means the device and control function that detects guard position; “carton blank” means a flat, pre-glued carton before erection; “change part” means removable tooling dedicated to a product or carton format; and “home position” means the defined machine-coordinate reference of a servo axis.

Do not translate equipment tags or identifiers such as Q0, B2, YV12, X14, A205, GA-AXP420-001. Preserve decimal values, units, warning severity, and the distinction between shall, must, should, and may.

The nominal tightening torque for the M8 stainless-steel guard fasteners is **24 N·m** (dry thread). For lubricated threads, use **18 N·m**.

COMMENT_TARGET: Verify that the customer-specific carton drawing and the machine recipe use the same dimensional reference convention before site acceptance testing.

| The obsolete instruction ~~TRACK_OLD~~ minimum pneumatic pressure is 5.5 bar requires engineering review before release.

10 Declaration and document status

This manual forms part of the machine technical documentation. It does not replace the machine-specific electrical schematic, pneumatic schematic, risk assessment, or declaration supplied with the equipment. In case of conflict, the serial-number-specific engineering documents take precedence over generic examples in this manual.

Generated sample company and machine: Northstar Packaging Systems S.r.l., Via Industriale 42, 25100 Brescia, Italy. All names, addresses, serial numbers, and technical values in this stress-test document are fictional.